

Why Spark came?

companies used hadoop successfully
HDFS → storage

MapReduce → processing

But, after some years

MapReduce is very slow.



mapReduce processes data by continuously reading from and writing to disk.

Eg:- [100 GB data] → MapReduce

Read from disk → process

Read again ← write to disk ←

↳ process again → write again

..... ←

Disk is slow ↗

What is memory?

Storage type	Speed
Hard disk	slow
RAM	fast

Note

Spark uses In-memory processing.



processing data directly in RAM instead of repeatedly reading/writing to disk.

What is Apache Spark

Apache Spark is a distributed data processing engine that processes large amounts of data quickly using in memory processing.

→ Apache : Organization that manages the project

→ Spark : processing engine.

→ Distributed : Uses multiple machines.

→ In-memory : Uses RAM.

Spark is a tool

- ↳ processes huge data
- ↳ Uses many machines
- ↳ Uses RAM for speed.

Hadoop

Storage + Processing
Uses MapReduce
Slower

Disk Based

Spark

Processing Engine
Uses Spark Engine
Faster

Memory Based

Why was spark created

Spark was created to process large data faster than MapReduce using in-memory processing.

Why is Spark faster than MapReduce?

Spark keeps data in memory whenever possible, reducing disk read and write operations.

Does spark load all data into one machines memory?

No, spark splits data into partitions and distributes them across multiple machines.

What is a cluster?

A cluster is a group of machines working together as one system.

What is partitioning?

process of splitting a large dataset into smaller chunks for parallel processing

Is a partition the same as a server?

No, a partition is a chunk of data while a server/executor processes the partition.

Can 1 executor process multiple partitions

Yes, 1 executor can process multiple partitions.

What is spill to disk?

Spill to disk occurs when Spark writes temporary data to disk because memory is insufficient.

Intermediate computation data is usually spilled to disk.

What is Driver?

The Driver receives code, creates execution plans, assigns tasks and coordinates execution.

What are executors?

Executors are worker processes that execute tasks and process data partitions.

Flow

code

Driver coordinates the work
Executors process the data

↓
Driver

plan creation

↓
Tasks

↓
Executors

↓
Results

↓
Driver

↓
Output

code

↓
Driver (build the plan)

when action is called

Spark creates tasks

Executors start processing partitions

when action is called

↓
Executors (evaluates the plan)

↓
Runs the task

What is lazy evaluation

Spark delays execution until an action is called

What is a transformation

Transformation creates a new dataset and builds the execution plan.

Give example of transformation

filter(), select(), join(), groupBy(),
withColumn()

What is an Action?

An action triggers the execution of the Spark job.

show(), count(), collect(), first(),
write()

Does Spark execute transformations immediately?

Spark only builds the plan until an action is called.

What is RDD (Resilient Distributed Dataset)
Spark's Original distributed data structure.

It doesn't contain schema information
(col names/types)

Harder to optimize → Spark doesn't know the structure of the data.

What is a Dataframe?

Distributed collection of data organized into named columns.

DF provide schema information and better optimization.

(col-name/dtypes/data) ✓

What is Query Optimization?

Finding the most efficient way to execute a query.

Column pruning?

Reads only the required columns and skips unnecessary columns.

It helps to reduce data reading, memory usage, Execution time.

Spark Planning

→ logical plan : Describes what operations need to be performed.

→ Optimized logical plan : logical plan after Spark applies optimizations

→ What is a physical plan : Describes how Spark will execute the job

⇒ Optimized logical plan will be used to generate the physical plan

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→ What is a physical plan : Describes how Spark will execute the job

⇒ Optimized logical plan includes column pruning

What is narrow transformation?

A transformation where each output partition depends on only one input partition.

`filter()`, `select()`, `withColumn()`, `drop()`

It doesn't require Shuffle.

What is wide transformation?

A transformation where output partitions depends on multiple input partitions

`groupBy()`, `join()`, `distinct()`, `orderBy()`, `repartition()`

What is Shuffle

Movement of data b/w partitions to bring related records together

It is expensive : it involves network communication & disk operations

Wide transformations usually cause shuffle

What is task?

Smallest unit of work given to the Executor

NO-of-tasks = NO-of-partitions

Executors [NO-of-core/CPU/...] $\times$ no-of-partitions
processing